



QuadPath: A Four-Expert Branch Architecture for Surgical Organ Segmentation

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Abstract

Segmentation of laparoscopic surgical images presents unique challenges due to tool occlusions, bleeding, deformable organ surfaces, specular reflections, and visual similarity between anatomical structures. Prior research by Guo et al. introduced a three-expert branch model for laparoscopic image segmentation that kept feature encodings mostly independent until the final fusion stage. This design allowed each convolutional expert branch to focus on different types of visual information, such as texture, boundary shape, and curved anatomical structures. Although the original model showed promising results, some organ classes still had low segmentation accuracy, with mIoU scores as low as 21% for difficult structures. These limitations suggested that the model could benefit from an additional expert branch capable of capturing broader contextual information. In this project, we extend the original TriPath architecture [1] by adding a MiT-B0 Vision Transformer branch, creating a QuadPath model for multi-organ surgical image segmentation. The goal of this added branch is to improve segmentation mask quality by combining local convolutional features with transformer-based global feature representations.

The Innovation

Laparoscopic surgical image segmentation is difficult because organs can have similar colors and textures, and the scene may include tool occlusions, bleeding, lighting changes, and specular reflections. This project focused on still images from the DSAD surgical dataset and used single-organ binary segmentation masks to evaluate how well each organ could be separated from the background. Prior TriPath results showed that some organs were segmented well, while smaller or visually complex structures had much lower accuracy. This raised the main research question:

Will adding a pretrained ViT as a global context expert improve surgical image segmentation accuracy?

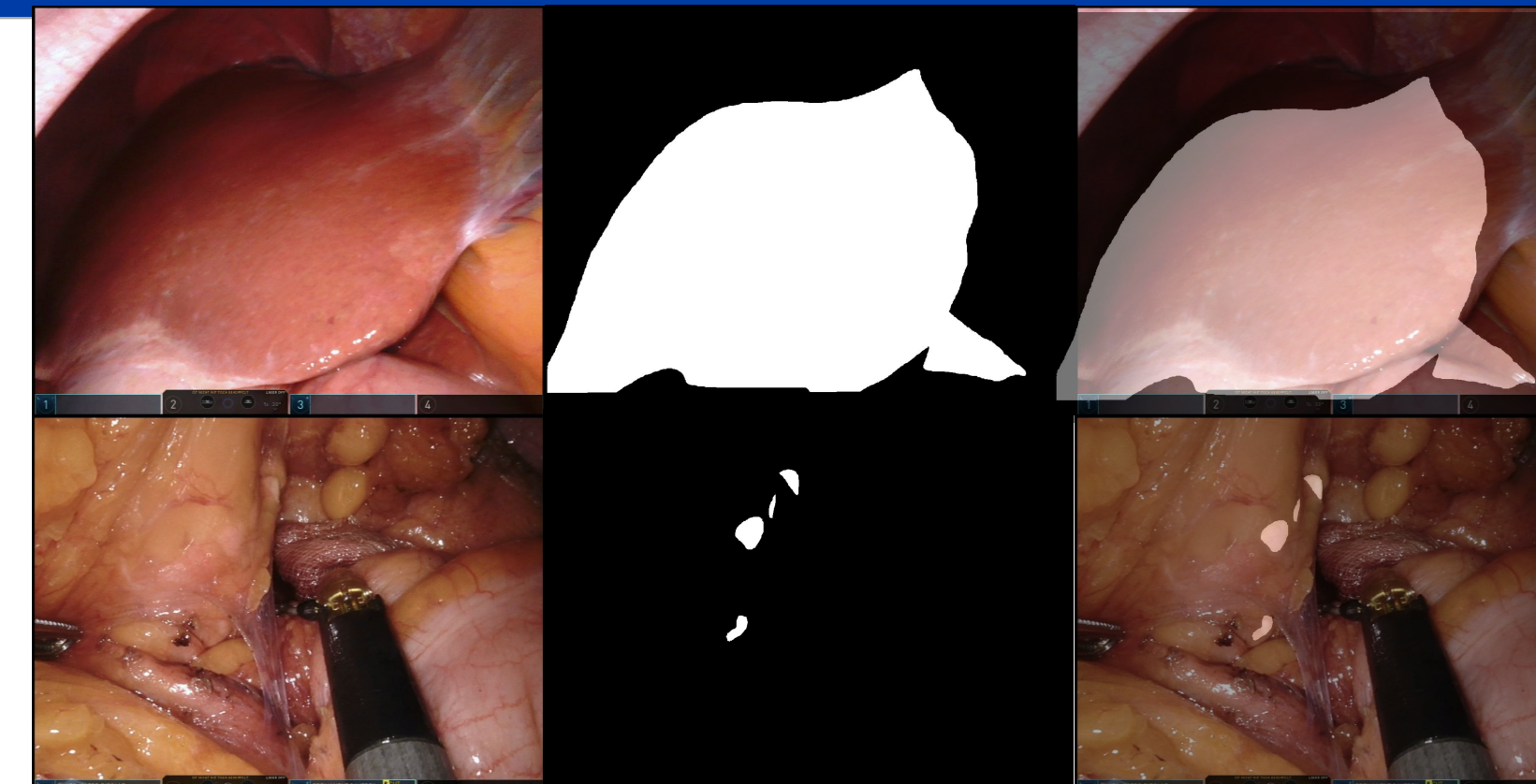
Dataset

This project used the DSAD surgical dataset, which contains laparoscopic surgical images with annotated organ segmentation masks. The dataset includes multiple abdominal organs such as the abdominal wall, colon, liver, pancreas, spleen, stomach, ureter, vesicular glands, and other surgical structures. For this project, the dataset was organized into single-organ binary segmentation tasks, where each model learned to separate one target organ from the background. This setup allowed us to evaluate how well the QuadPath model performed across both larger organs and smaller, more difficult structures. Each image-mask pair was preprocessed before training by resizing the images, matching the masks to the corresponding organ class, and normalizing the image inputs for compatibility with the pretrained MiT-B0 branch. This allowed the model to combine the DSAD surgical image data with transformer-based feature extraction.

Da Vinci Surgical System

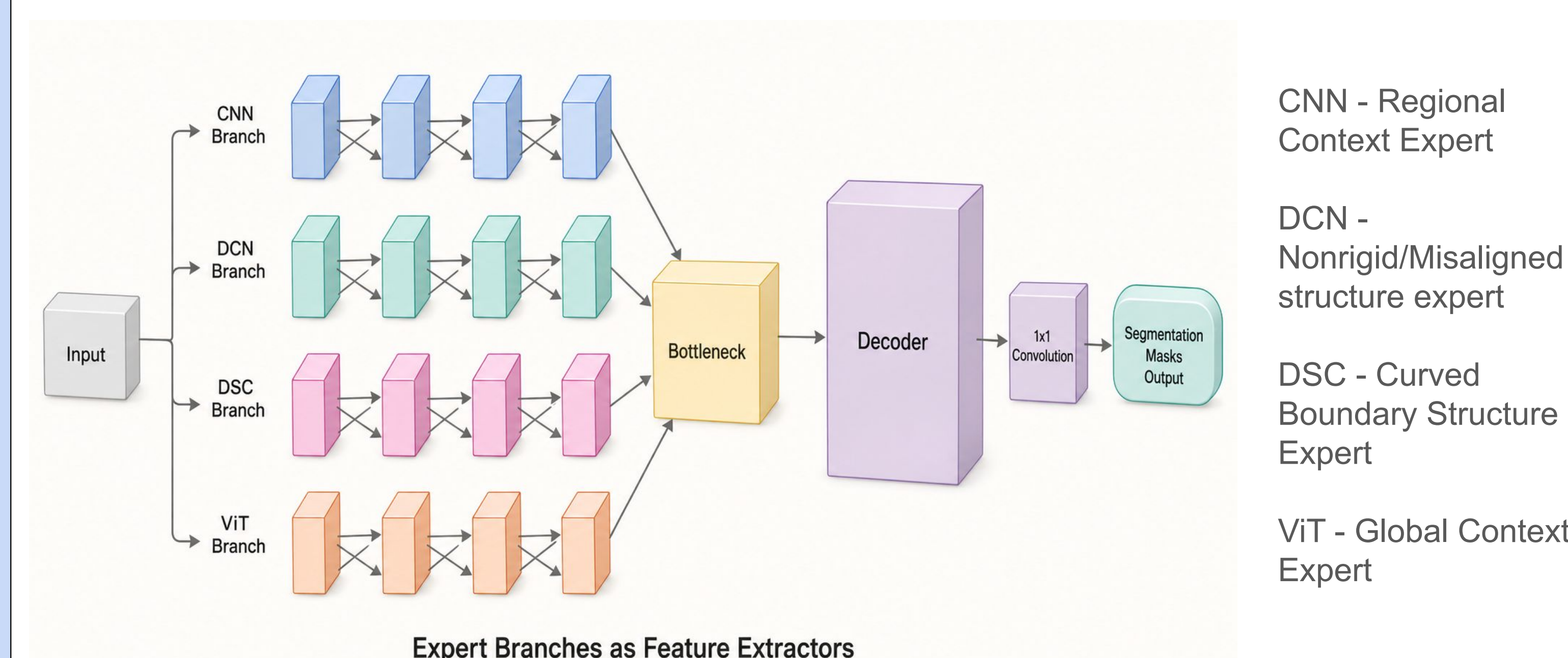


Model of usage of the da vinci robotic surgical system for laparoscopic minimally invasive procedures.



Sample image of a liver (top) and pancreas (bottom) and the ground truth binary segmentation mask from the DSAD dataset.

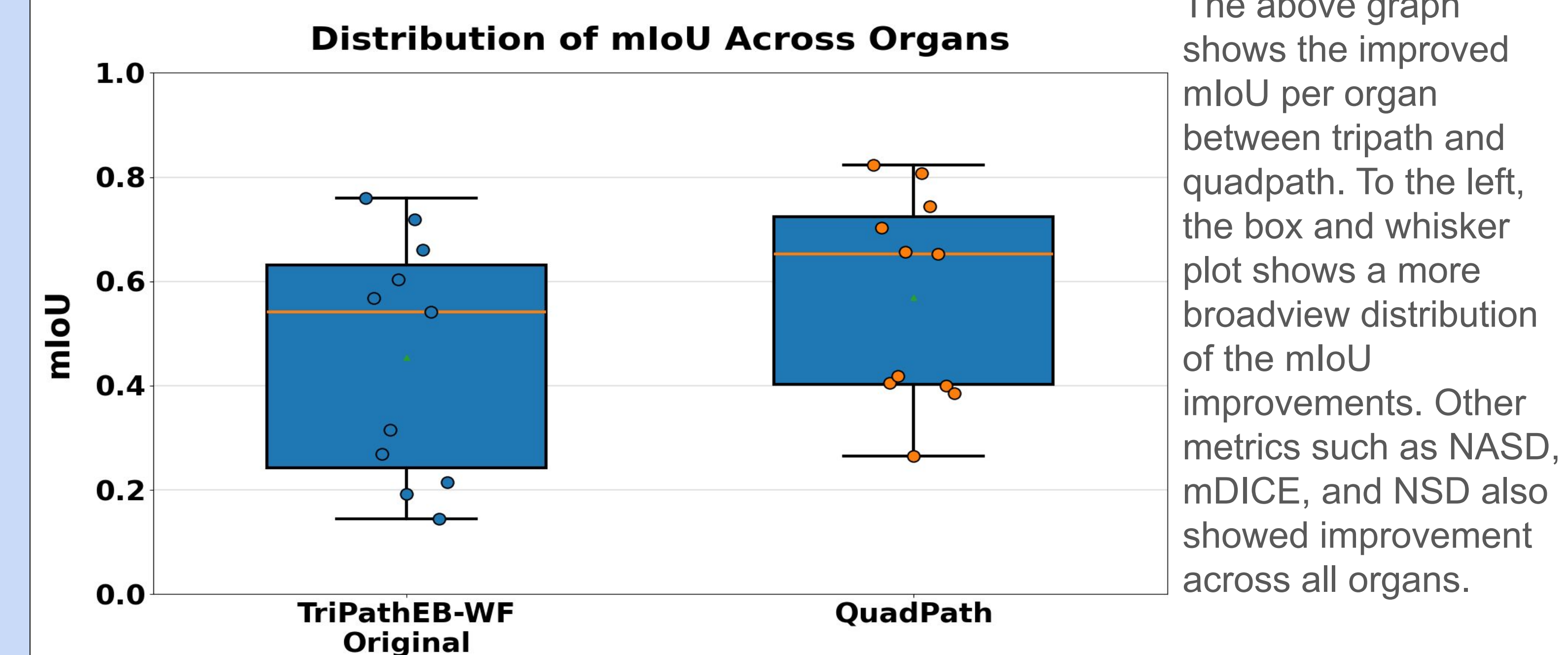
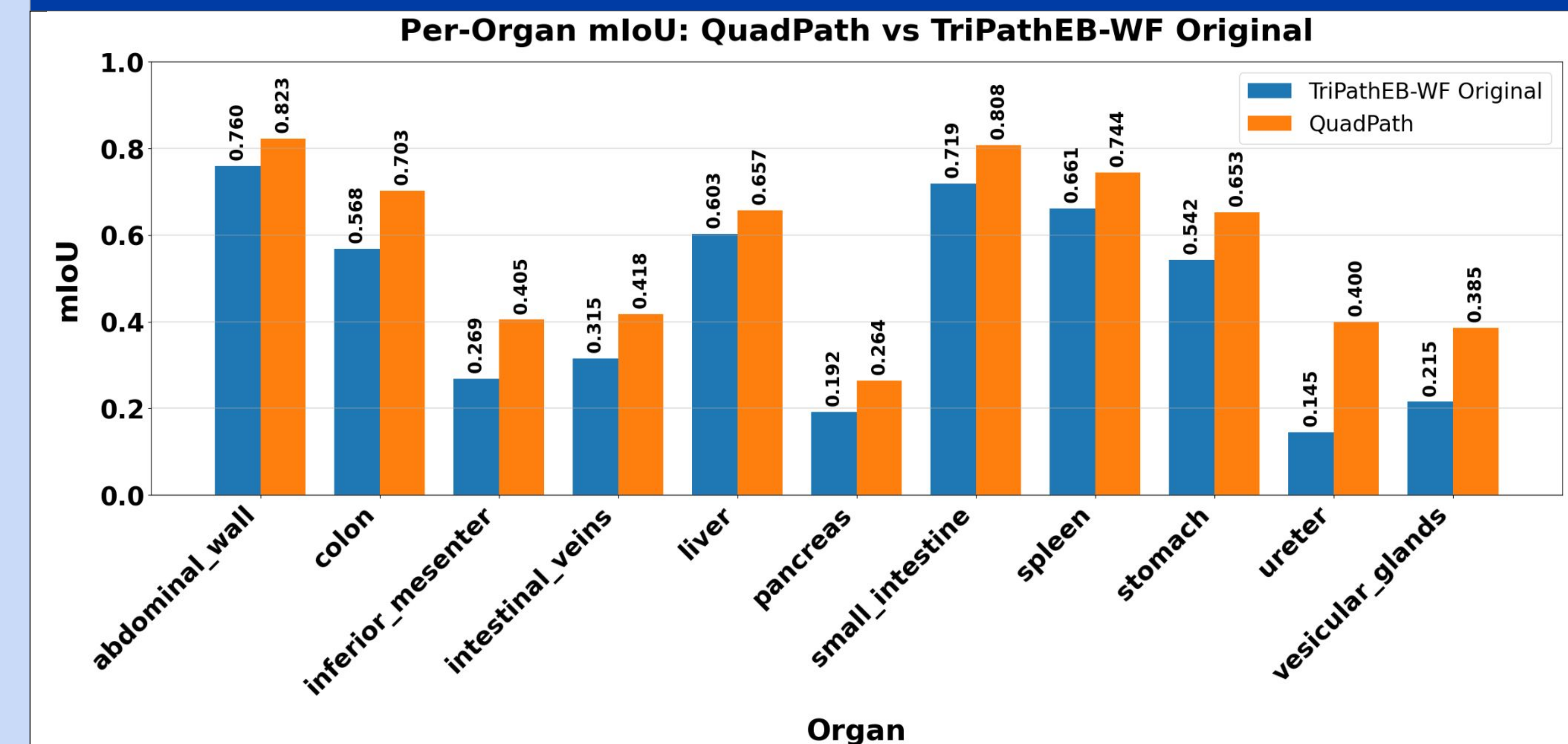
QuadPath Neural Network Architecture



Methods

For this project, the original TriPath segmentation model was extended by adding a fourth expert branch based on the MiT-B0 Vision Transformer encoder. MiT-B0 was selected because it provides transformer-based global feature extraction while still being lightweight enough for surgical image segmentation. The DSAD images and masks were organized by organ class, resized to match the model input requirements, and normalized using ImageNet mean and standard deviation so they were compatible with the pretrained MiT-B0 encoder. Mutli-scale features from the MiT-B0 branch were then projected using 1x1 convolution layers to align them with the UNet skip connection dimensions. These transformer features were combined with the existing CNN, DCN, and DSC branch features through skip connection fusion, allowing the decoder to use information from all four expert branches. The final quadpath model was evaluated using mIoU, mDice, MASD, and NSD, and its performance was compared against the original TriPathEB-WF baseline as well as some sample results from ViT running independently on the dataset.

Results



Conclusion

Adding the MiT-B0 transformer branch improved the original TriPathEB-WF model across the main evaluation metrics. The QuadPath model achieved higher average mIoU, mDice, and NSD, while also reducing MASD, showing better overlap accuracy and boundary quality. The largest mIoU improvements were seen in harder organ classes such as the ureter, vesicular glands, inferior mesenteric artery, and pancreas. This suggests that the added transformer branch helped the model capture broader context and improve segmentation for smaller or more difficult structures. Overall, the results show that combining CNN, DCN, DSC, and transformer-based features can strengthen multi-organ surgical segmentation compared to the original TriPath model. Future work should further test the model on more cases and compare additional metrics beyond mIoU to better evaluate multi-organ detection performance.

References

- [1] L. Guo, C. Camerota, M. Mahmoud, and F. Esposito, "Expert Branches: Module Diversity for Stronger Feature Learning in Laparoscopic Segmentation," *Proceedings of Machine Learning Research*, MIDL 2026, 2026.